

Changing connections: temporal potential structural
connectivity, unprotected pinch points, and monitoring
priorities for free-ranging cheetahs (*Acinonyx jubatus*)
in southern Africa

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September 2, 2026

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Abstract

[TODO: Context: cheetah range contraction, majority of range outside protected areas.]

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased drastically in the past decades. Using a multivariate analysis, with the core variables being fixed at the historical baseline, so that the temporal signal is landscape-side. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas. This leaves "barriers" like ranchers, roads and other human development in areas where cheetahs will be roaming.

[TODO: Gap: existing regional work models habitat suitability, not observed change in network permeability; the 2026 KAZA multispecies assessment excluded cheetah outright.]

Models looking at habitat suitability have been done in the past, but very little predictive modeling of the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright.

[TODO: What we did: held range cores fixed at the 2010–2016 confirmed-range baseline and modelled potential structural connectivity across five predeclared resistance scenarios at four snapshots spanning 2012–2024.] [TODO: Headline result: magnitude and geography of connectivity loss.] [TODO: Second result: share of robust pinch points outside protected areas.] [TODO: Third result: ranked monitoring cells and their stability across weightings.] [TODO: Claim boundary sentence: these are modelled structural priorities, not demonstrated cheetah movement.]

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

Move 1: Why cheetah connectivity, and why outside protected areas

The southern African cheetah has had their range substantially reduced, with their current range estimated to be on majority unprotected land. Cheetah range on protected land is governed by centralized park-management initiatives [TODO: Maybe get an example of that?]. Unprotected land poses a different type of challenge, with landowners, local communities and governmental entities involved. This paper looks at the landscape connectivity change between 2012 and 2024, but also provides a predictive look at what future land permeability for cheetah movement may be based on three scenarios: conservation of the land/low-growth, continuation, and high-development. The predictive scenarios are based on the assumption that the core range of cheetahs is fixed at the 2010-2016 baseline, and that any change in permeability is due to landscape changes rather than changes in the core range itself. Which is to say, the core range is not expected to change in the future, but the landscape surrounding it may become more or less permeable to cheetah movement. Based on a multivariate analysis, the results of this study show that the southern African cheetah is at risk of further range contraction and isolation due to anthropogenic development, and that conservation efforts should focus on maintaining and enhancing connectivity between core range areas.

Move 2: What is already modelled, and what is not

[TODO: Habitat suitability for this species and region is now well covered, including current and projected suitability with protected-area mismatch. State plainly that this project does not add another suitability model. Then the pivot: suitability is not permeability, and no published work reports how permeability between established cheetah range cores has *changed*. Note that the recent regional multispecies connectivity assessment excluded cheetah for insufficient data, which is simultaneously the justification and the constraint.]

Move 3: The methodological frame

[TODO: Least-cost and circuit-theoretic connectivity (Adriaensen et al., 2003; McRae et al., 2008); resistance parameterisation is the dominant source of uncertainty in this class of model (Zeller et al., 2012, 2018); hence a predeclared scenario ensemble rather than a single preferred surface. Graph-level importance follows Saura and Pascual-Hortal (2007); metric choice follows Keeley et al. (2021). Design choices are framed against Beier et al. (2008).]

Move 4 — Objectives and hypotheses

[TODO: State the three objectives (change, protection gap, monitoring priority) and list H1–H4 with their directional predictions. Every hypothesis must be falsifiable as written; if you cannot describe the result that would refute it in one clause, rewrite it. See the Hypotheses sheet.]

Terminology. Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a cell of high modelled current concentration, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range plus a buffered analysis extent. All layers were aligned to a fixed equal-area projection and a 1 km snap grid, so every raster is directly comparable cell for cell. Cores were derived from the historical confirmed-range baseline and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs.

2.2 Rationale for the working resolution

We used a 1-km working resolution because it represents a compromise between the coarse 10-km occurrence baseline and the finer environmental covariates, which range from 30 m to 1 km. This resolution is fine enough to identify meaningful pinch points in the landscape while avoiding the implication of precision that the occurrence evidence cannot support.

2.3 Range cores

[TODO: WF-011. Three or more core definitions from the archived confirmed-range data, differing in evidence-category threshold and minimum patch area. Report how many cores each definition yields. State explicitly that cores are HELD CONSTANT across all four temporal snapshots, and why (CAU-002): no comparable range reassessment exists after the baseline period, so any core movement would be invented.]

We defined a core area as a contiguous patch of cells that were confirmed presence in the historical baseline. Cells were grouped into patches using an 8-neighbor rule, and only patches larger than a minimum area threshold were retained. I used three core definitions from the archived confirmed-range data, differing in evidence-category threshold and minimum patch area. The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the absence of comparable range reassessment.

[OUTLINE]

We represented the 23 baseline-defined core areas as fixed nodes across all four temporal snapshots. Candidate links were defined as the union of an undirected three-nearest-neighbor graph based on Euclidean distances between core centroids and the graph's minimum spanning tree. Removing duplicate undirected pairs yielded 45 selected core pairs. These straight-line links defined the pairs evaluated using least-cost analysis and were not interpreted as observed movement corridors.

2.4 Covariates and their temporal treatment

[TODO: Table 1 lists every layer, its native resolution, its source year(s), and critically whether it is treated as DYNAMIC (varies across snapshots) or STATIC (held fixed, contributing no temporal signal). Do not bury this distinction. Roads, livestock, terrain and hydrography are static; built-up fraction, nighttime radiance and continuous vegetation fields are dynamic.]

[OUTLINE]

- Built-up fraction: GHSL, 2012 / 2016 / 2020 / 2024.
- Nighttime lights: VIIRS, flare-masked before differencing.
- Land cover: MODIS MCD12Q1, reclassified to open vegetation, woody, cropland, built, water, bare.
- Vegetation continuous fields (VCF): tree cover, non-tree vegetation, bare ground.
- Roads: GRIP4 Africa, clipped to the study extent.
- Livestock: GLW4 cattle, goats and sheep, combined into a livestock-exposure index.
- Terrain: approximately 30 m SRTM DEM pulled through Google Earth Engine and aggregated to the 1 km grid. Derivatives: mean slope, maximum slope, terrain resistance surface.
- Veterinary fences: WWF/KAZA reference fence dataset, with main, near-barrier and conservative fence scenarios. Kruger fence references treated as scenario information, not as universal barriers.
- Dynamic: built-up fraction, nighttime lights, tree cover, non-tree vegetation, bare-ground VCF.
- Static: roads, livestock, terrain, hydrography, fences, protected areas, core definitions.
- Categorical land cover used for context and masking only, never as an independent temporal resistance penalty.

Why categorical land cover is not the change carrier. [TODO: CAU-008 and the MODIS class-instability problem. Annual categorical products flip classes between years for reasons unrelated to landscape change. Temporal signal is therefore carried on continuous fields; categorical land cover is used for masking and description only.]

2.5 Resistance scenarios

[TODO: WF-013, WF-014. Present RES-01 to RES-05 as a table of variable-to-resistance functions with the evidence or rationale for each. State that the scenario set was fixed before results were viewed and that no single scenario is designated as preferred. Report the transformation (linear vs exponential) per scenario and cite Zeller et al. (2012) for why the transformation matters as much as the weights.]

[OUTLINE]

Terrain component. The DEM was not used directly as resistance; slope was derived from it. Mean slope carries the primary terrain surface, maximum slope retained for sensitivity and context. Terrain resistance function: slope $\leq 10^\circ$ low resistance; 10 to 30° linearly increasing; $\geq 30^\circ$ capped high; resistance rescaled to 1 to 10.

Original primary temporal surface. Anthropogenic component 80% of total, terrain 20%. Within the anthropogenic component: built-up 30%, roads 30%, livestock 30%, nighttime lights 10%.

Three internal weighting models compared: primary, equal, infrastructure-emphasis.

Vegetation correction. The first temporal model omitted VCF from the resistance calculation, which is a mismatch with the predeclared protocol. Three further sensitivity scenarios were added, keeping the original anthropogenic-only surfaces intact rather than replacing them:

- vegetation-low: 10% vegetation, 70% anthropogenic, 20% terrain
- vegetation-balanced: 20% vegetation, 60% anthropogenic, 20% terrain
- vegetation-high: 40% vegetation, 40% anthropogenic, 20% terrain

The vegetation proxy uses continuous VCF fields to represent sparse and bare cover. It does not claim to measure fine-scale habitat edge or observed movement preference, and should be described that way.

Reconciliation still needed: these six weighting models are not the same object as the predeclared RES-01 to RES-06 scenario set. Either map each one onto a predeclared scenario ID or state plainly in the text that the implemented set differs and why.

2.6 Connectivity modelling

[TODO: WF-015, WF-016. Least-cost / effective-distance work in ArcGIS Pro (Distance Accumulation, Optimal Region Connections, Corridor). Current flow computed with Circuitscape.jl (Anantharaman et al., 2020) in pairwise mode; give the solver, the source/ground configuration, and how current was normalised for comparability across years. State why the omnidirectional approach (Landau et al., 2021) was NOT used: cores are defined a priori, so a core-to-core formulation is the correct one.]

[OUTLINE]

- Least-cost paths computed between all 45 selected core pairs.
- Cost-distance analysis run on a common snap raster, cell size, mask, extent and projection, so surfaces are comparable across years and scenarios.
- Implementation shortcut worth documenting: one **CostDistance** run per source core, then multiple destination paths extracted from that single accumulated surface with **CostPathAsPolyline**. Equivalent result to pairwise runs, avoids recomputing the same source surface 45 times. Say it is equivalent and say why.
- Circuitscape current-flow stage not yet run. Either report it once complete or change this subsection to describe a least-cost-only design.

Corridor definition. [TODO: Decide and state ONCE whether corridors are (a) continuous current surfaces summarised by quantile or (b) thresholded polygons. This choice sets the denominator for the protected-area comparison in §2.9 and cannot be made after seeing the overlap result.]

2.7 Network structure

[TODO: WF-017. Cores as nodes, effective distances as edge weights. Report the probability of connectivity and per-node dPC (Saura & Pascual-Hortal, 2007), computed in Conefor (Saura & Torné, 2009) or an equivalent, and betweenness on the same graph. Justify the metric set against Keeley et al. (2021). Report sensitivity to the dispersal-distance parameter; do not report a single value.]

[OUTLINE]

edge betweenness computed on the 23-core network, link importance ranked on normalised edge betweenness. PC and dPC not yet computed, and dispersal distance is still the 100 km placeholder.

2.8 Temporal comparison

[TODO: WF-018. Same cores, same scenarios, same resolution across all four snapshots, so that differences are attributable to covariate change alone. Give the change equations. Report both absolute and proportional change.]

[OUTLINE]

Completed snapshot reports exist for 2012, 2016, 2020 and 2024. First-pass anthropogenic-only comparison:

- all 45 links present in every year
- all 32 robust links have both 2012 and 2024 costs
- median robust-link change approximately -0.002%

- largest robust-link increase approximately 0.349%
- no robust link changed by $\geq 1\%$

This is not the final temporal result, because that model omitted VCF. The vegetation-weight sensitivity run is in progress and supersedes it. Write the subsection so the near-zero change finding is framed as a real result about modelled permeability rather than a null to be explained away, and note that it holds under a model with no vegetation term.

2.9 Protected-area coverage

[TODO: WF-019, CAU-011, CAU-012. Name the WDPA <MONTH YEAR> release. State polygon-only handling and what was excluded. State the null explicitly: high-current area is compared against [TODO: the full extent OR the modelled network] using an area-standardised permutation test. Predeclare this.]

2.10 Robustness and uncertainty

[TODO: WF-022. The full sensitivity grid: five resistance scenarios $\times$ core definitions $\times$ resolutions $\times$ snapshots. Report the resulting run count. Define "robust" numerically (e.g. a cell in the top decile of current under at least k of n runs) and fix k in advance. Produce an agreement map. Low-agreement locations are reported as uncertain, never as priorities.]

[OUTLINE]

- 90 weight comparisons evaluated.
- 34 of 45 core pairs robust under the tested alternative weights, 11 sensitive.
- Fence scenarios changed 5 links: 1–13, 1–17, 9–13, 13–17, 27–30.
- 32 links classified primary-priority eligible; 13 retained as uncertainty and data-collection priorities.

Two things to resolve before writing: state the numeric robustness rule actually applied and whether it matches the predeclared $k/n = 0.8$; and reconcile the 34 robust pairs with the 32 primary-priority-eligible links, since those are different counts and a reader will notice.

2.11 Independent plausibility assessment

[TODO: WF-020, WF-021, CAU-004, CAU-017. See the validation decision note before writing this. If the plausibility-check route is taken, this subsection is titled as above and must NOT use the word "validation". If the empirical-resistance route is taken, add RES-06, describe the fitting procedure, and report country-held-out performance following Roberts et al. (2017).]

2.12 Monitoring prioritisation

[TODO: WF-023, WF-024, CAU-020. Components, normalisation, and the predeclared weight set, plus at least two alternative weightings. Report rank stability (Spearman / Kendall) across weightings. Publish component scores alongside the composite so a reader can reweight.]

[OUTLINE]

link ranking currently rests on normalised edge betweenness plus the robust / sensitive split, with 32 primary-priority-eligible and 13 uncertainty-priority links. The full MCDA with three weight sets is not yet built.

2.13 Software and reproducibility

[TODO: ArcGIS Pro [TODO: version]; Circuitscape.jl [TODO: version]; Julia [TODO: version]; R [TODO: version] or Python [TODO: version]; Google Earth Engine for acquisition and harmonisation. All geoprocessing scripted in arcpy rather than ModelBuilder. Archive location and what can and cannot be redistributed (CAU-012, CAU-013).]

[OUTLINE]

ArcGIS Pro Spatial Analyst via arcpy (`CostDistance`, `CostPathAsPolyline`); Google Earth Engine for SRTM, GHSL, VIIRS, MODIS MCD12Q1 and VCF acquisition and harmonisation. Record versions.

3 Results

3.1 Connectivity change, 2012–2024

[TODO: H1. Magnitude and geography of change in current flow and effective distance. Report per country and per core-pair. Figure 1.]

3.2 Robust pinch points

[TODO: H3. How many locations met the robustness criterion, and how concentrated they are. Figure 2, Table 2.]

3.3 Protected-area coverage of priority connectivity

[TODO: H2. Observed share outside protected areas versus the predeclared null, with the permutation result. Table 3.]

3.4 Model sensitivity

[TODO: Agreement across the sensitivity grid. This subsection should be substantial. If corridor rankings are unstable, say so here plainly; the Discussion then pivots to the uncertainty-guided monitoring framing rather than a corridor map. Figure 3.]

3.5 Independent plausibility assessment

[TODO: Outcome of §2.11. Report it even if weak or mixed.]

3.6 Monitoring priorities

[TODO: H4. Ranked cells and their stability. Figure 4, Table 4.]

Figure 1: Change in modelled current flow between snapshots, 2012–2024. [TODO: Colour-blind-safe diverging ramp; uncertainty shown, not hidden.]

Figure 2: Robust pinch points and their protected-area status.

Figure 3: Ensemble agreement across the sensitivity grid.

4 Discussion

What the results mean, at the level the evidence supports

[TODO: First paragraph restates the headline finding in one sentence and immediately bounds it. Resist writing "cheetah movement" anywhere in this section.]

4.1 Where permeability has declined, and what co-occurred

[TODO: H1. Frame as spatial co-occurrence, not causation (CAU-014). If the dominant pressure differs by country, report the heterogeneity rather than compressing it into a single anthropogenic narrative.]

4.2 The protection gap as a governance question

[TODO: H2. If the outside-PA share is high, the defensible reading is that connectivity for this species depends on land under other management, which is a statement about where conservation effort must operate, not evidence that protected areas have failed. Note the confound that protected areas are often sited in land of low agricultural value.]

Figure 4: Ranked monitoring priority cells.

Table 1: Covariates, native resolution, source years, and static/dynamic treatment.

4.3 Robust versus model-sensitive linkages

[TODO: H3. This is the paper’s most transferable contribution. Robust locations support action; model-sensitive locations define the survey agenda. Note the caveat that scenarios sharing input layers can agree artificially.]

4.4 From uncertainty to a survey design

[TODO: H4. The monitoring index is decision support under stated assumptions, not an objective ranking (CAU-020). Say what a partner organisation would actually do with it, and acknowledge that accessibility and cost will reorder the list in practice.]

4.5 Comparison with regional connectivity work

[TODO: Position against the elephant and multispecies transfrontier assessments (Brennan et al., 2020; Naidoo et al., 2024) as methodological neighbours, and against the fencing-effects literature (Osipova et al., 2018). Be explicit that those are different species with different movement ecology: the methods transfer, the resistance values do not (CAU-015).]

5 Limitations

Fixed historical cores. [TODO: CAU-002. The single most important limitation. Cores derive from 2010–2016 evidence and do not move; all temporal signal is landscape-side.]

Structural, not functional. [TODO: CAU-001.]

Table 2: Top 20 robust pinch points with country, protection status, and agreement score.

Table 3: Protected-area coverage of high-current area against the predeclared null.

Table 4: Monitoring priority ranking with component scores under alternative weightings.

Sampling bias in the occurrence baseline. [TODO: CAU-003. Absence of records is not evidence of absence.]

Resistance is expert-assigned. [TODO: CAU-005. What the scenario ensemble does and does not compensate for.]

Static covariates. [TODO: Roads, livestock, fences and terrain contribute no temporal signal; consequences for interpreting change.]

Mapping completeness varies by country. [TODO: CAU-010.]

Geographic transferability. [TODO: CAU-015.]

Scope. [TODO: CAU-018. Climate, conflict and occupancy are deliberately out.]

6 Conclusion

[TODO: Three or four sentences. One on what changed, one on where the robust priorities are and their protection status, one on the monitoring recommendation, one closing the claim boundary. No new information, no hedging language you have not already earned.]

Data availability

[TODO: CAU-012, CAU-013. Derived outputs and all code are archived at [TODO: OSF/Zenodo DOI]. Raw protected-area data are not redistributed and must be obtained from Protected

Planet under its terms. Occurrence data are shared only at the resolution permitted by the source archive.]

Acknowledgements

[TODO: Department, any data providers, any partner organisation that reviewed the monitoring priorities.]

Conflicts of interest

The author declares no conflicts of interest.

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A Analysis protocol as predeclared

[TODO: Paste the locked protocol document here verbatim, with its version and date. This is what lets you write "predeclared" in the Methods without overclaiming. It should include: extent and exclusion rules, full CRS WKT, working resolution and the alternatives to be tested, core definitions, the resistance scenario table, the sensitivity grid, the robustness threshold k , the MCDA weight sets, and the claim dictionary listing prohibited phrasings.]

B Supplementary material

[TODO: Temporal alignment table (CAU-007); correlation/VIF screening among human pressure covariates (CAU-016); per-scenario current maps; per-resolution results; full monitoring ranking with component scores; software environment manifest.]